

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	G. Santosh Kumar	Reg. No.:	192524020
Section / Batch:	2025 Batch	Date:	10/07/2026

Code of Conduct

I G. Santosh Kumar _____ (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Santosh Kumar _____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
GRAND TOTAL:			100 Marks

CO1_AT1_C CODE OUTPUT PREDICTION

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QUESTIONS

Q1

C Code – Array Traversal
10 marks

Q2

C Code – Linear Search
10 marks

Q3

C Code – Binary Search Mid
10 marks

Q4

C Code – Pointer with Array
10 marks

Q5

C Code – Array Update
10 marks

Q6

C Code – Linked List Node
10 marks

Q7

C Code – Time Complexity Loop
10 marks

Q1 C Code – Array Traversal

10 marks

```
#include <stdio.h>
int main() {
    int arr[] = {5, 10, 15, 20};
    int i, sum = 0;
    for(i = 0; i < 4; i++)
        sum += arr[i];
    printf("%d", sum);
    return 0;
}
```

Your Answer

50

[Save Answer](#)

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QUESTIONS

Q1

C Code – Array Traversal
10 marks

Q2

C Code – Linear Search
10 marks

Q3

C Code – Binary Search Mid
10 marks

Q4

C Code – Pointer with Array
10 marks

Q5

C Code – Array Update
10 marks

Q6

C Code – Linked List Node
10 marks

Q7

C Code – Time Complexity Loop
10 marks

Q10 C Code - Pointer Increment

10 marks

```
#include<stdio.h>
int main(){
    int a[]={5,10,15};
    int *p=a;
    p++;
    printf("%d",*p);
}
```

Your Answer

10

[Save Answer](#)

My Assignments

Course: CSA0305RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) OPEN

DEBUGGING & OPTIMIZATION TASK

0/10 answered

[Attempt →](#)

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) OPEN

DEBUGGING & OPTIMIZATION TASK

0/10 answered

[Attempt →](#)

CO3_AT3_MCQ OPEN

Multiple Choice Questions - Non Linear Data Structures

0/25 answered

[Attempt →](#)

CO1_AT1_C CODE OUTPUT PREDICTION OPEN Completed

C CODE OUTPUT PREDICTION

10/10 answered

[Review →](#)